

A Comparative Study of Classical and Transformer-Based Models for Phishing Email Detection

Bachelor Semester Project 6

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ABSTRACT

The project was developed by student **João Bernardo Sousa Faria**, under the supervision of supervisor **Salima Lamsiyah**.

Phishing emails remain a major cybersecurity threat, since they attempt to deceive users into revealing sensitive information, clicking malicious links, or trusting fraudulent requests. The project studies phishing email detection as a binary natural language processing task, where each email is classified as either safe or phishing. The objective is to **compare a classical feature-based model**, based on **TF-IDF and Logistic Regression**, with a transformer-based model based on **fine-tuned BERT**.

The primary dataset was cleaned, deduplicated, and split into stratified training, validation, and test sets. The models were evaluated using accuracy, phishing-class precision, phishing-class recall, phishing F1-score, macro-F1, micro-F1, and weighted-F1. Decision thresholds were selected on the validation set and final results were reported on the held-out test set. The best classical model achieved 0.9814 accuracy and 0.9752 phishing F1-score, while BERT achieved 0.9875 accuracy and 0.9832 phishing F1-score. However, the difference between BERT and the combination of TF-IDF and Logistic Regression was not very large, showing that the classical baseline remains highly competitive. A second-dataset experiment using the DIFraud phishing benchmark showed that performance decreases when models are evaluated across datasets, suggesting that phishing

detection models are sensitive to dataset-specific language patterns.

INTRODUCTION

Phishing emails are a common and dangerous form of cyberattack. They are designed to trick users into opening malicious links, downloading malware, revealing passwords, or trusting fraudulent requests. Since email is still widely used in personal, academic, and professional communication, automatic phishing detection is an important task in cybersecurity.

From a machine learning perspective, phishing email detection can be treated as a binary text classification problem. Given the content of an email, the system predicts whether the email is legitimate or malicious, which makes the task closely related to natural language processing, since the model must learn textual patterns such as suspicious vocabulary, urgency, unusual formatting, links, promotional language, or requests for action.

The project investigates the question “**How do classical feature-based machine learning models compare with transformer-based models in phishing email detection tasks?**”

The comparison is used for highlighting differences between classical models and transformer models. TF-IDF combined with Logistic Regression is simple, fast, and interpretable. BERT, on the other hand, can model contextual relationships between words, but requires more computational resources and is more difficult to interpret.

The main contributions of the project are:

- A complete phishing email classification pipeline
- comparison between TF-IDF + Logistic Regression and BERT
- validation-based threshold tuning
- an error and explainability analysis
- a cross-dataset experiment to test generalization.

RELATED WORK

Phishing email detection is commonly treated as a supervised text classification problem. Classical approaches often rely on feature extraction methods such as TF-IDF combined with machine learning classifiers such as Logistic Regression, Support Vector Machines, or Random Forests. These methods are efficient and interpretable because they represent emails through weighted terms and allow for the inspection of important features.

More recent approaches use transformer-based language models such as BERT, which are based on the Transformer architecture [3]. These models are pre-trained on large text corpora and can be fine-tuned for downstream classification tasks. Their advantage is that they produce contextual representations, meaning that the interpretation of a word depends on the surrounding words. This is useful for phishing detection, where malicious intent can depend on phrasing, sentence structure, or context rather than only isolated keywords.

Transformer models also have disadvantages such as the requirement for more computational resources. They are also slower to train and less directly interpretable than classical linear models. For this reason, the project compares both approaches and includes explainability and cross-dataset experiments to better understand the behavior of the models beyond accuracy alone.

SCIENTIFIC BACKGROUND

Binary Text Classification

Binary text classification assigns one of two labels to each document. In the project, label **0** represents a **safe email** and label **1** represents a **phishing email**.

Since phishing emails are the security-relevant class, phishing precision, recall, and F1-score are especially important.

TF-IDF and Logistic Regression

TF-IDF transforms **text** into **numerical vectors** by giving importance to words that are frequent in a document but less frequent across the full corpus. Logistic Regression then learns a **linear decision threshold** over these features. With this, the learned coefficients can be inspected to understand which terms push predictions toward the phishing or safe class.

BERT Transformer-Based Classification

BERT follows a different approach as a model pre-trained for contextual language representation [4]. It produces **contextual token representations**, meaning that the representation of each token depends on the words around it. It is beneficial when phishing intent depends on sentence structure or wording rather than only individual keywords.

However, BERT is more computationally expensive, requires fine-tuning, and is less transparent than Logistic Regression. Therefore, comparing BERT with a strong classical baseline gives a more balanced understanding of performance, complexity, and interpretability.

Evaluation Metrics

The models are evaluated using several metrics. Accuracy measures the overall proportion of correct predictions, but it can be misleading when the classes are not perfectly balanced. For this reason, the report also focuses on precision, recall, and F1-score for the phishing class.

Phishing precision measures how many emails predicted as phishing are truly phishing. Phishing recall measures how many real phishing emails are correctly detected. The phishing F1-score combines precision and recall into a single value. Phishing F1-score is fundamental, due to phishing detection requiring balance between **missing dangerous phishing emails** and **incorrectly flagging safe emails**.

Macro-F1, micro-F1, and weighted-F1 are also reported. Macro-F1 gives equal importance to both classes, while micro-F1 measures global classification performance. weighted-F1 accounts for the class distribution. Collectively, these metrics provide a more complete view of model behavior than accuracy alone.

METHODOLOGY

The primary dataset used in the project is “**zefang-liu/phishing-email-dataset**” from Hugging Face [1]. It contains email texts and labels indicating whether each message is a safe email or a phishing email. The original dataset contained 18,650 rows in the notebook environment. The preprocessing step kept only the email text and label columns, removed missing values, mapped safe emails to 0 and phishing emails to 1, renamed the text column to text, and removed duplicate email texts. Deduplication was performed before splitting the dataset, since duplicated emails can cause data leakage if the same message appears in both training and test sets. After cleaning and deduplication, 17,537 emails remained. The dataset contained 10,979 safe emails and 6,558 phishing emails. Table 1 summarizes the datasets used in the experiments.

Dataset	Rows	Safe/Benign	Phishing	Phishing %
Primary phishing email dataset	17,537	10,979	6,558	37.40%
DIFraud phishing	15,272	9,198	6,074	39.77%

Table 1. Dataset statistics after preprocessing

The cleaned dataset was split into **training**, **validation**, and **test** sets with the use of stratified sampling with a fixed random state. The split used 70% of the data for training, 15% for validation, and 15% for testing. Stratification preserved the class distribution across the three splits. The validation set was used for threshold tuning and, in the BERT experiment, checkpoint selection. The test set was kept unseen until final evaluation. Three main models were evaluated. The first model was **TF-IDF + Logistic Regression**, implemented with Scikit-learn [5]. It used English stop-word removal, a maximum of 5,000 TF-IDF features, and validation-based threshold tuning. The second model

used the same TF-IDF representation but applied **balanced class weights** to Logistic Regression. The third model was **BERT**, using bert-base-uncased, fine-tuned for binary sequence classification. The BERT model was fine-tuned using the HuggingFace Transformers library [6] for two epochs with a learning rate of 2×10^{-5} , batch size 8, and weight decay 0.01. Emails were tokenized with padding and truncation to a maximum length of 128 tokens. A token-length analysis showed that many emails were longer than this limit, so truncation is an important limitation of the experiment. For all models, the decision threshold was tuned on the validation set. Instead of always using the default threshold of 0.50, thresholds between 0.05 and 0.95 were tested, and the threshold maximizing phishing F1-score on the validation set was selected. The selected threshold was then applied once to the test set. A second-dataset experiment was also performed using the phishing domain of the DIFraud benchmark [2]. This experiment tested whether a model trained on one dataset could generalize to another. The cross-dataset comparison used TF-IDF + Logistic Regression, since it is efficient and interpretable.

Implementation Details beyond the Initial Declaration

Implementation details beyond the initial declaration were also added in the notebook. In addition to the basic comparison between TF-IDF + Logistic Regression and BERT, the implementation includes a **class-weighted Logistic Regression variant** to study the effect of class imbalance. It also performs validation-based threshold tuning instead of relying only on the default 0.50 decision threshold, due to phishing detection being sensitive to the balance between false positives and false negatives.

The notebook also includes **explainability analysis** for the classical model. With Logistic Regression being linear, its coefficients can be inspected to identify terms that push predictions toward the phishing or safe class. An instance-level explanation function was also implemented to show which TF-IDF terms contributed most to a specific prediction.

This makes the classical model useful not only as a baseline but also as a diagnostic tool.

A **zero-shot Natural Language Inference baseline** was also implemented. This experiment uses a pre-trained model without task-specific fine-tuning to classify emails as safe or phishing. It provides an exploratory comparison between task-specific training and general-purpose language model reasoning.

To improve reproducibility, the notebook uses a **fixed random seed** for data splitting and model training. The final results are saved into CSV files, including the main model comparison and the cross-dataset comparison. This makes the experiments easier to verify and reuse.

EXPERIMENTAL RESULTS

The final test-set results show that the supervised models performed strongly on the primary dataset. The standard TF-IDF + Logistic Regression model achieved **0.9814 accuracy** and **0.9752 phishing F1-score**. The class-weighted Logistic Regression model achieved the same phishing F1-score of **0.9752**, with slightly higher phishing recall but slightly lower precision. BERT achieved the **best overall performance**, with **0.9875 accuracy** and **0.9832 phishing F1-score**. Table 2 presents the final test-set results for the main models. Additionally, zero-shot NLI was evaluated on a stratified subset of 200 test emails due to its higher computational cost.

MODEL	THRESHOLD	ACCURACY	PHISHING PRECISION	PHISHING RECALL	PHISHING F1-SCORE	MACRO F1-SCORE	MICRO F1-SCORE	WEIGHTED F1-SCORE
TF-IDF + Logistic Regression	0.45	0.9814	0.9708	0.9797	0.9752	0.9801	0.9814	0.9814
TF-IDF + Logistic Regression (balanced)	0.54	0.9814	0.9698	0.9807	0.9752	0.9802	0.9814	0.9814
BERT	0.08	0.9875	0.9847	0.9817	0.9832	0.9866	0.9875	0.9875
Zero-shot NLI	N/A	0.6200	0.4615	0.0800	0.1364	0.4464	0.6200	0.5239

Table 2. Main model comparison on primary test set

The results show that BERT was the strongest model, but the improvement over the classical baseline was moderate. This suggests that many phishing cues in the primary dataset can already be captured by **lexical features**. The class-weighted Logistic Regression model did not significantly improve final F1-score, which suggests that the class imbalance was moderate

and already handled reasonably well by stratified splitting and threshold tuning.

The error analysis showed that TF-IDF + Logistic Regression misclassified more emails than BERT. Some phishing emails looked similar to legitimate newsletters or technical mailing-list messages, which made them difficult for both models. Some safe emails also contained suspicious-looking elements such as links, short requests, or unusual formatting. These examples show that phishing detection is not only a keyword-matching task. Some phishing emails use professional or technical language, while some safe emails contain features that look suspicious, which explains why contextual models such as BERT can improve performance, but also show why even BERT still misclassifies some examples.

The Logistic Regression model also provided a simple form of explainability through its coefficients. Positive coefficients pushed predictions toward phishing, while negative coefficients pushed predictions toward safe email, which allowed for the inspection of the terms that influenced the classical model. BERT was less directly interpretable, and future work could add explainability methods such as LIME, SHAP, Integrated Gradients, or token occlusion.

The cross-dataset experiment showed a clear generalization problem. Same-dataset performance was **strong for both datasets**, but cross-dataset transfer **reduced** performance. Table 3 reports the cross-dataset generalization results.

Experiment	Accuracy	Prec. phishing	Recall phishing	F1 phishing
Primary train/test	0.9814	0.9708	0.9797	0.9752
DIFraud train/test	0.9719	0.9879	0.9408	0.9638
Primary trained → DIFraud test	0.9051	0.8360	0.9474	0.8882
DIFraud trained → Primary test	0.7948	0.9703	0.4654	0.6291

Table 3. Cross-dataset generalization results

The primary-trained model transferred to DIFraud better than the DIFraud-trained model transferred to the primary dataset. The least positive result was obtained when the DIFraud-trained model was tested on the primary dataset, where phishing recall dropped strongly. This suggests that the two datasets contain different phishing patterns and that strong

performance on one dataset does **not** guarantee robust performance on another.

DISCUSSION AND LIMITATIONS

The main finding is that BERT achieved the **best** performance, but TF-IDF + Logistic Regression remained highly competitive. This result is important because **Logistic Regression is faster, simpler, and more interpretable**. In a real-world setting, a classical model could be useful as a lightweight baseline or screening system, while BERT could be used when **higher predictive performance** is required.

The results also show a trade-off between performance and complexity. BERT benefits from contextual representations, but it requires more computational resources and provides less transparent predictions. Logistic Regression cannot model context in the same way, but its coefficients are easy to inspect and explain.

Additionally, the cross-dataset experiment shows that the models may learn dataset-specific patterns. For example, a model may learn the writing style, formatting, or source-specific vocabulary of one dataset rather than general phishing behavior, which limits the reliability of models evaluated only on one dataset.

Regarding limitations, many are present in the project. For a start, BERT used a maximum input length of 128 tokens, even though many emails were longer. Important information may therefore have been truncated. Furthermore, only one transformer model was fine-tuned. Models such as RoBERTa, DistilBERT, or long-context transformers could give different results. Additionally, public phishing datasets may contain noisy or ambiguous labels, especially due to spam, scams, marketing, and phishing being capable of overlapping. Finally, the zero-shot baseline was implemented to explore whether a general-purpose Natural Language Inference model could classify phishing emails without task-specific fine-tuning. It was evaluated on a stratified subset of 200 test emails because inference was slower and more computationally expensive than for the trained models. The results showed that zero-shot NLI was not competitive with the supervised

models, especially because of its very low phishing recall.

CONCLUSION AND FUTURE WORK

The project compared classical and transformer-based models for phishing email detection. The implemented pipeline cleaned and deduplicated the data, preserved class balance with stratified splitting, tuned decision thresholds on a validation set, and evaluated final performance on a held-out test set. BERT achieved the best final results, with 0.9875 accuracy and 0.9832 phishing F1-score. However, TF-IDF + Logistic Regression was also very strong, achieving 0.9814 accuracy and 0.9752 phishing F1-score. This shows that classical feature-based models remain valuable for phishing email detection, especially when efficiency and interpretability matter. The second-dataset experiment showed that generalization remains a challenge. Models trained and tested on the same dataset performed positively, but performance dropped when tested across datasets, which suggests that future phishing detection systems should be evaluated on multiple sources before being considered reliable.

Future work could increase BERT's input length, split long emails into chunks, test additional transformer models, train on mixed datasets, apply domain adaptation, and add explainability methods for BERT. Deployment-oriented metrics such as false-negative rate, false-positive rate, latency, and probability calibration would also make the evaluation more realistic.

Future work could also include a more systematic zero-shot and few-shot evaluation.

ACKNOWLEDGMENTS

I would like to thank my supervisor **Salima Lamsiyah** for the guidance I have been given. Availability was agreed between both and acted upon when required. I was comfortable reaching out at any time when necessary and I am very satisfied with the environment I experienced throughout the entire project.

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